MCP: AI Apps as Easy as 1, 2, 3

INTRODUCTION TO MODEL CONTEXT PROTOCOL (MCP)

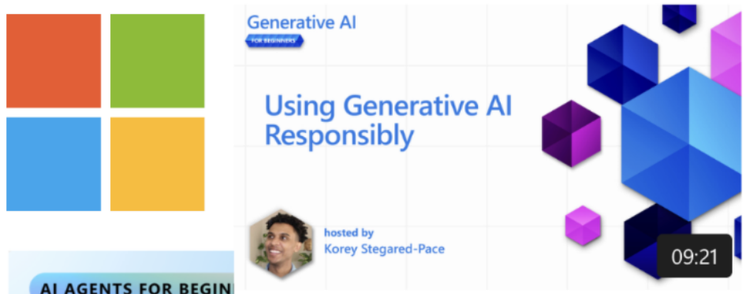
James Chapman
AI Curriculum Manager, DataCamp

datacamp

Meet Your Instructor

Korey Stegared-Pace

Senior AI Cloud Advocate at Microsoft



EPISODE

Using Generative AI Responsibly (Part 3 of 18) | Generative AI for Beginners

Jun 25, 2024

Azure

EPISODE

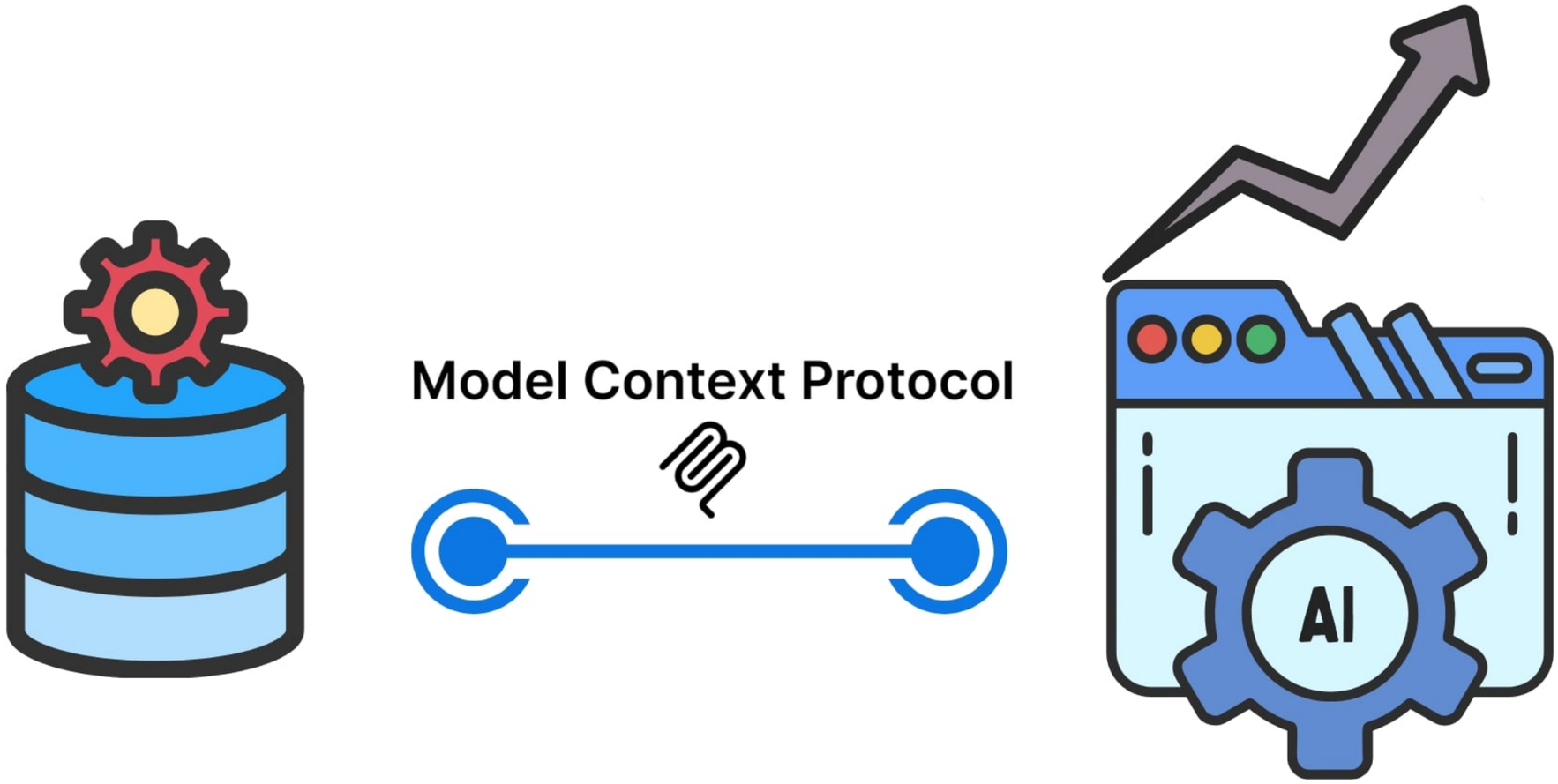
What are AI agents? (Part 1 of 10) | AI Agents for Beginners

Mar 24, 2025

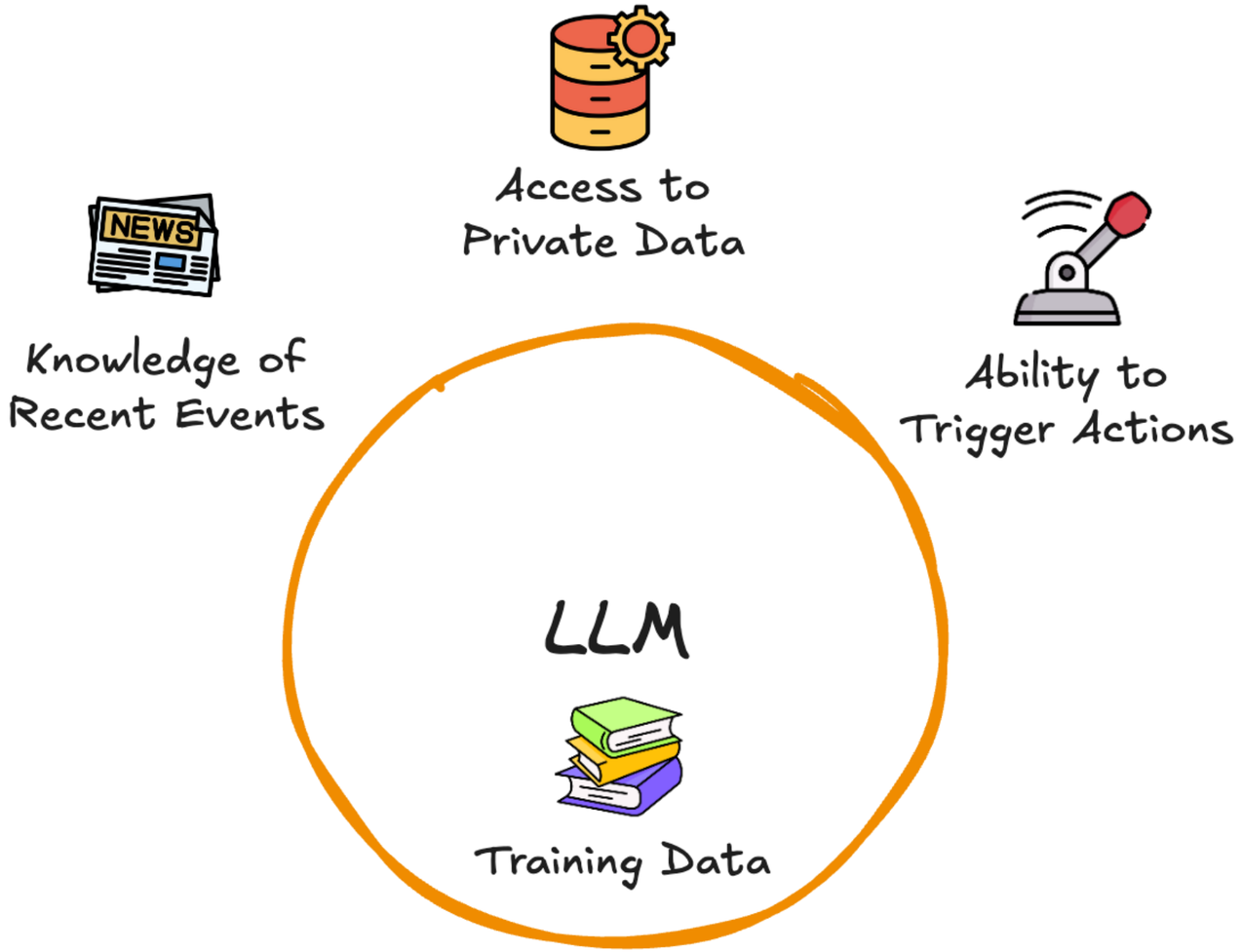
Beginner

[1] https://developer.microsoft.com/en-us/advocates/korey-stegared-pace

Looking Ahead



The Big LLM Problem



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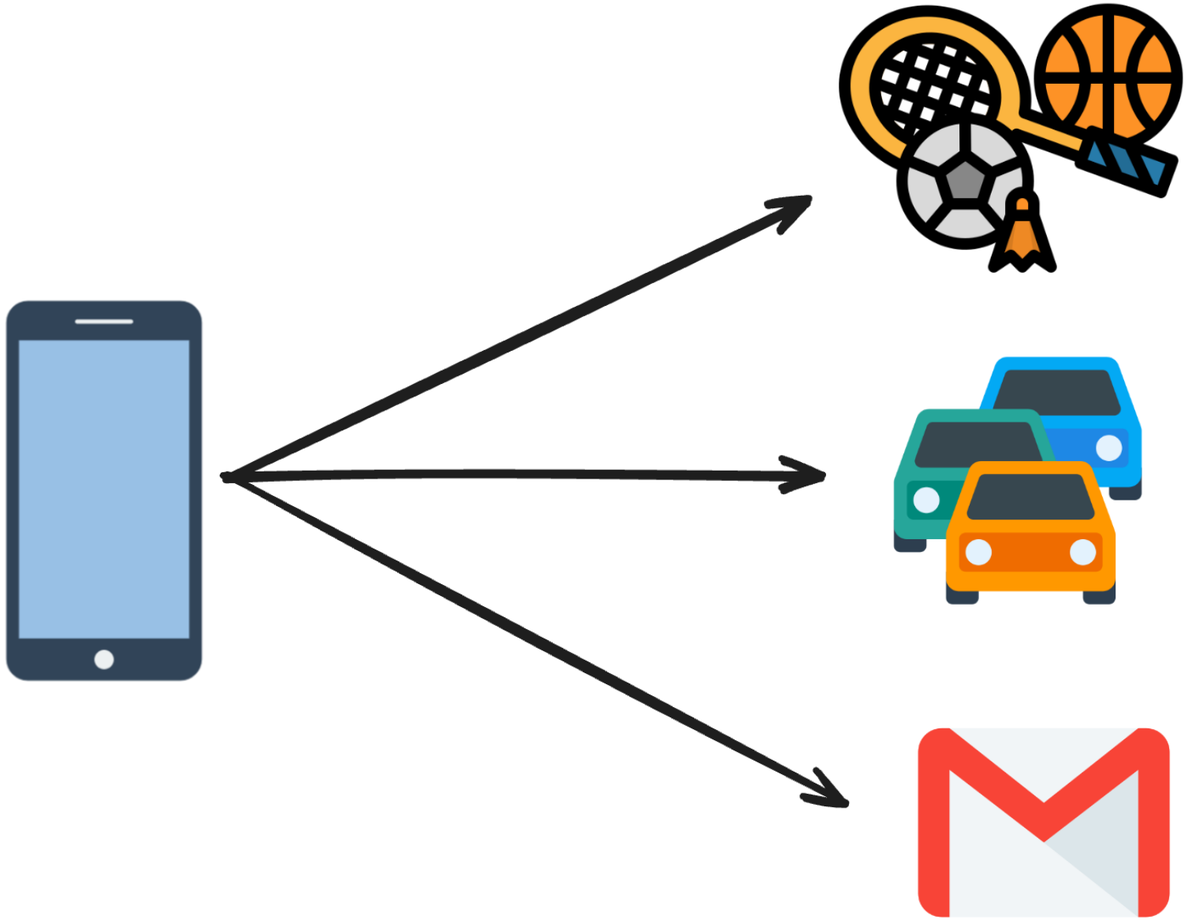

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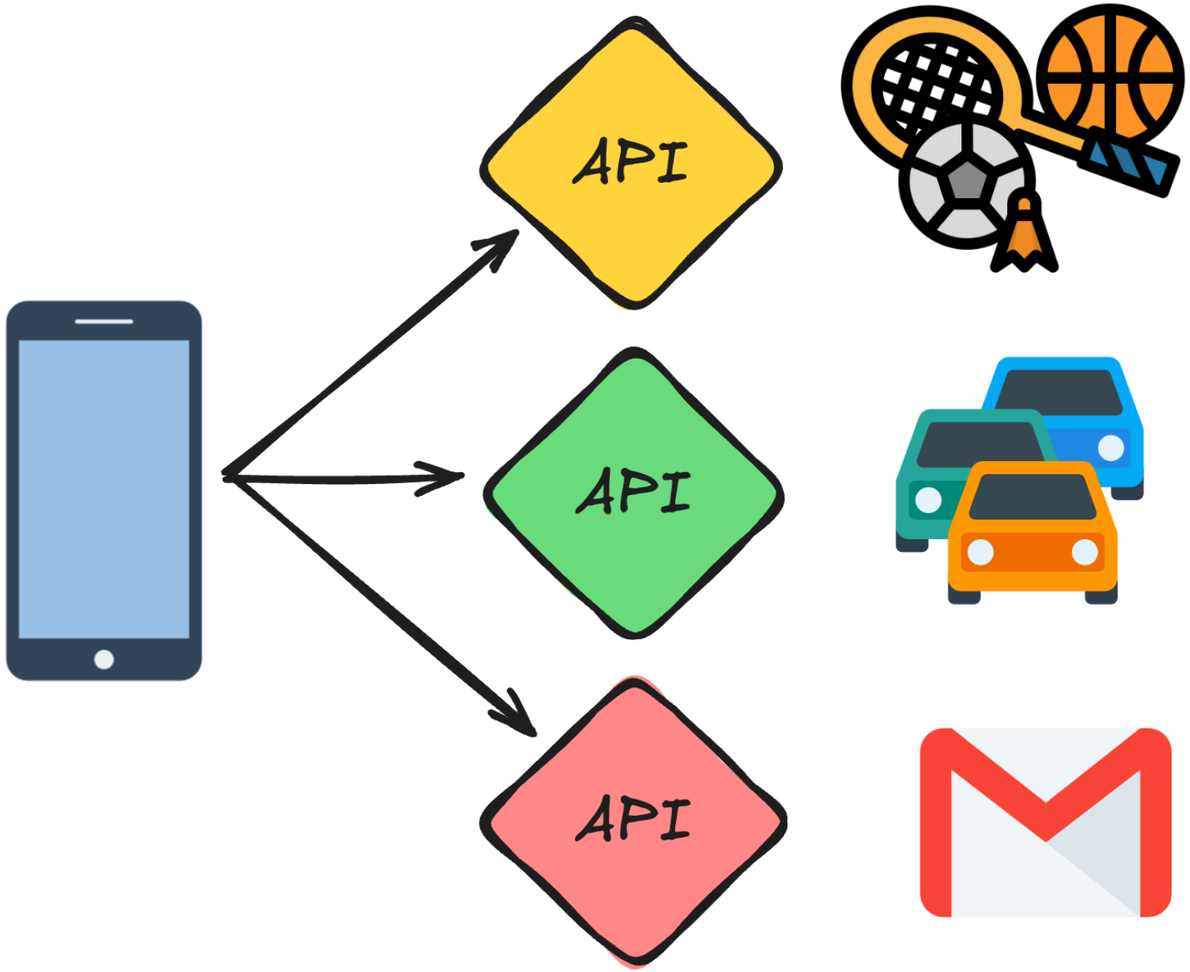

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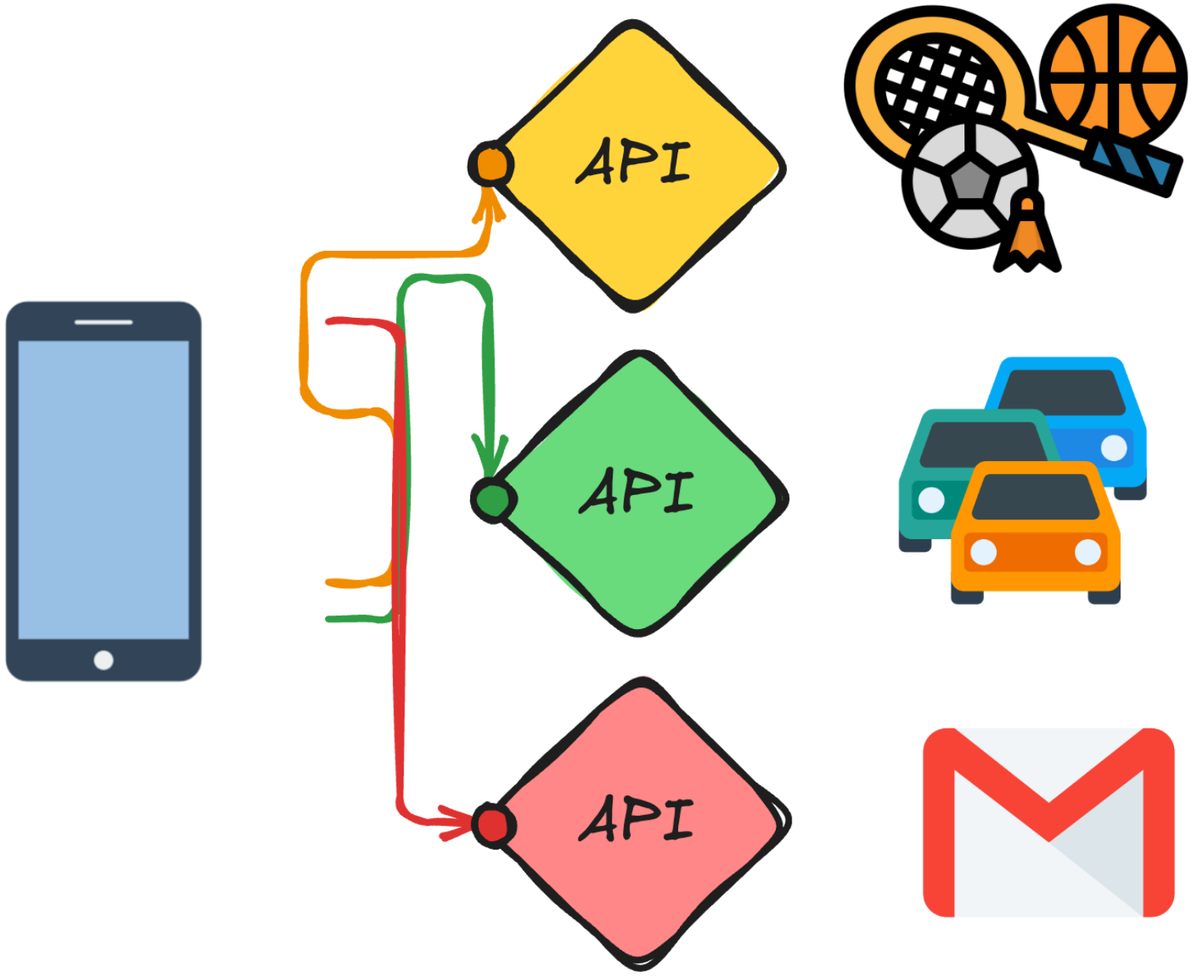
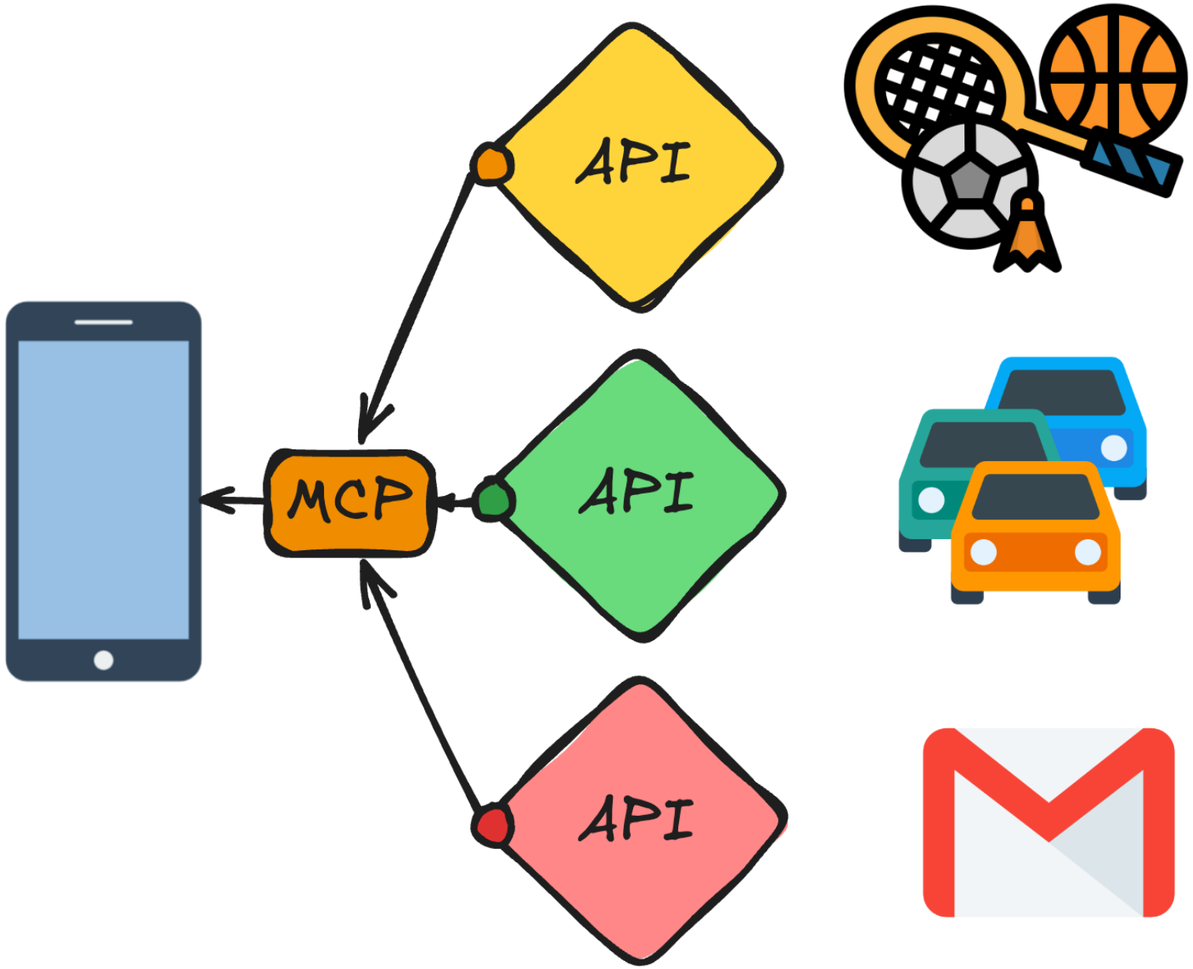


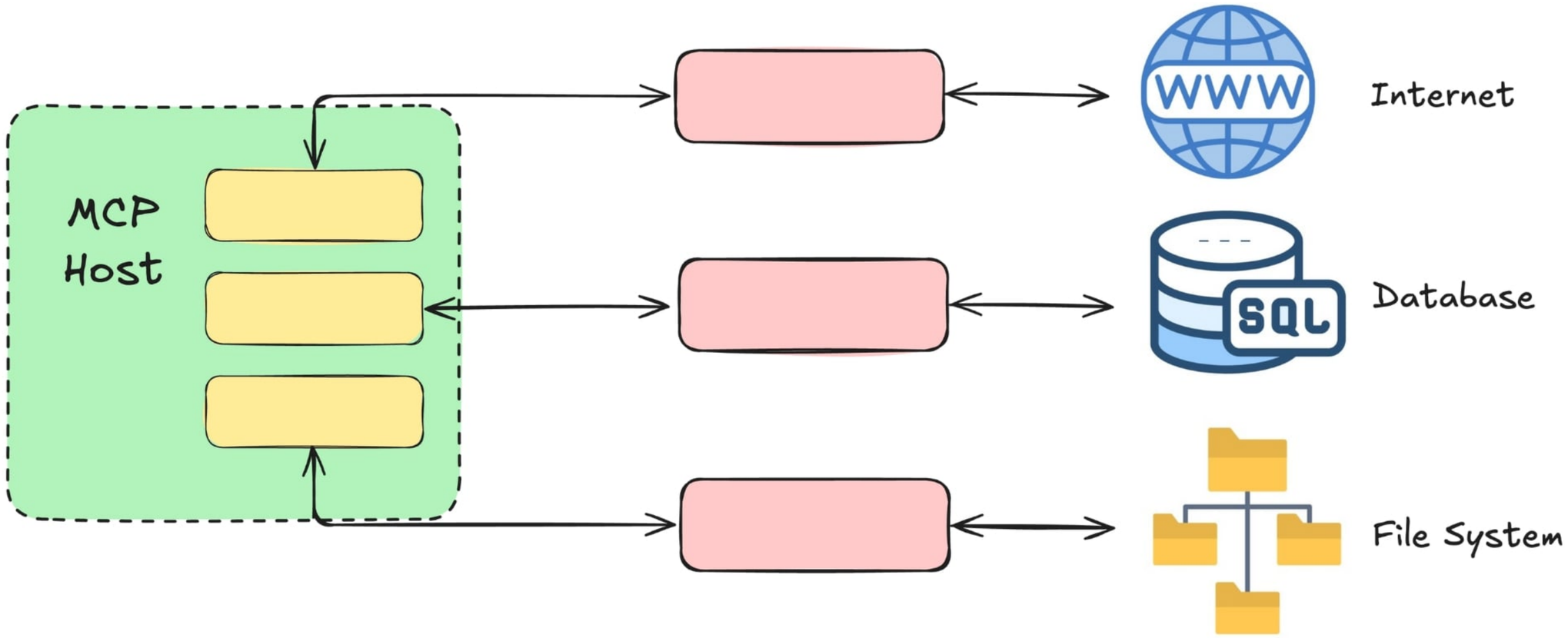
MCP to the Rescue!



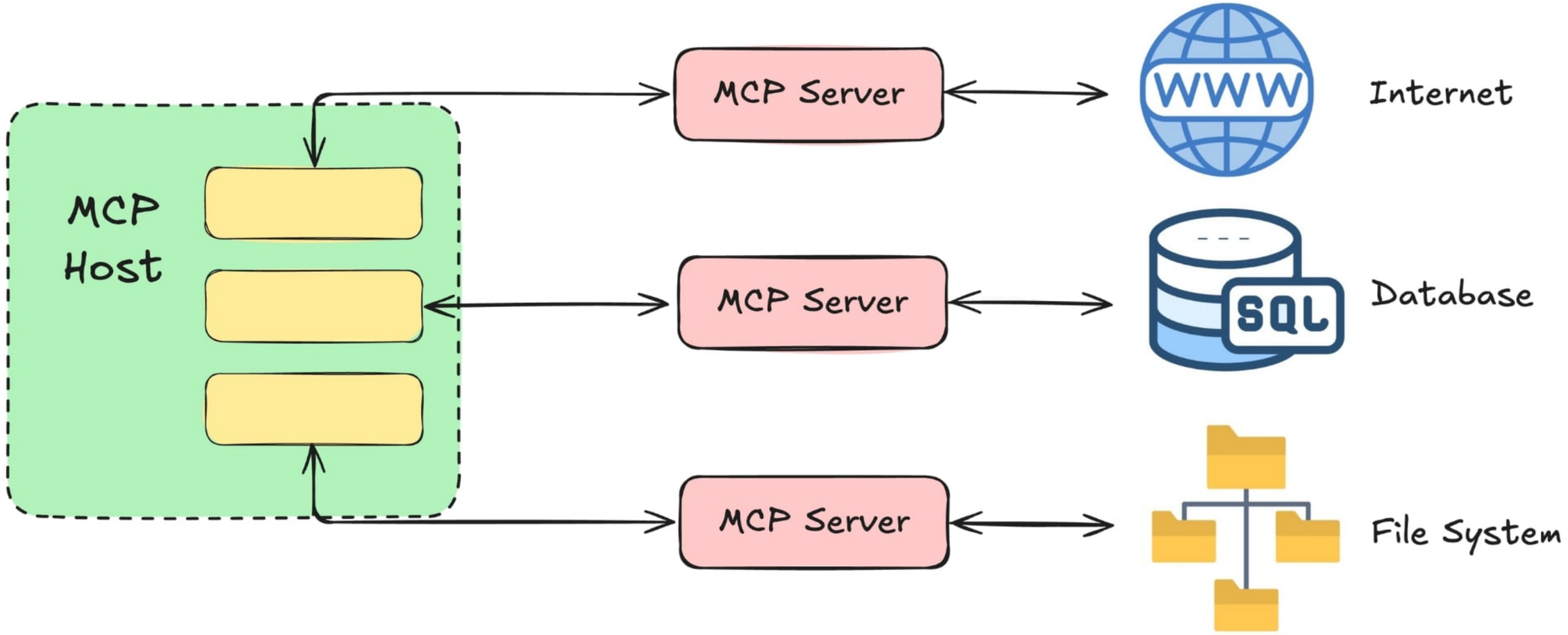
MCP to the Rescue!

- **Simplifies connectivity** and adds features
  - *Dynamic tool discovery* → AI can "see" what functionality is available
- MCP components:
  - Host, client, and the server

The MCP Architecture



The MCP Architecture



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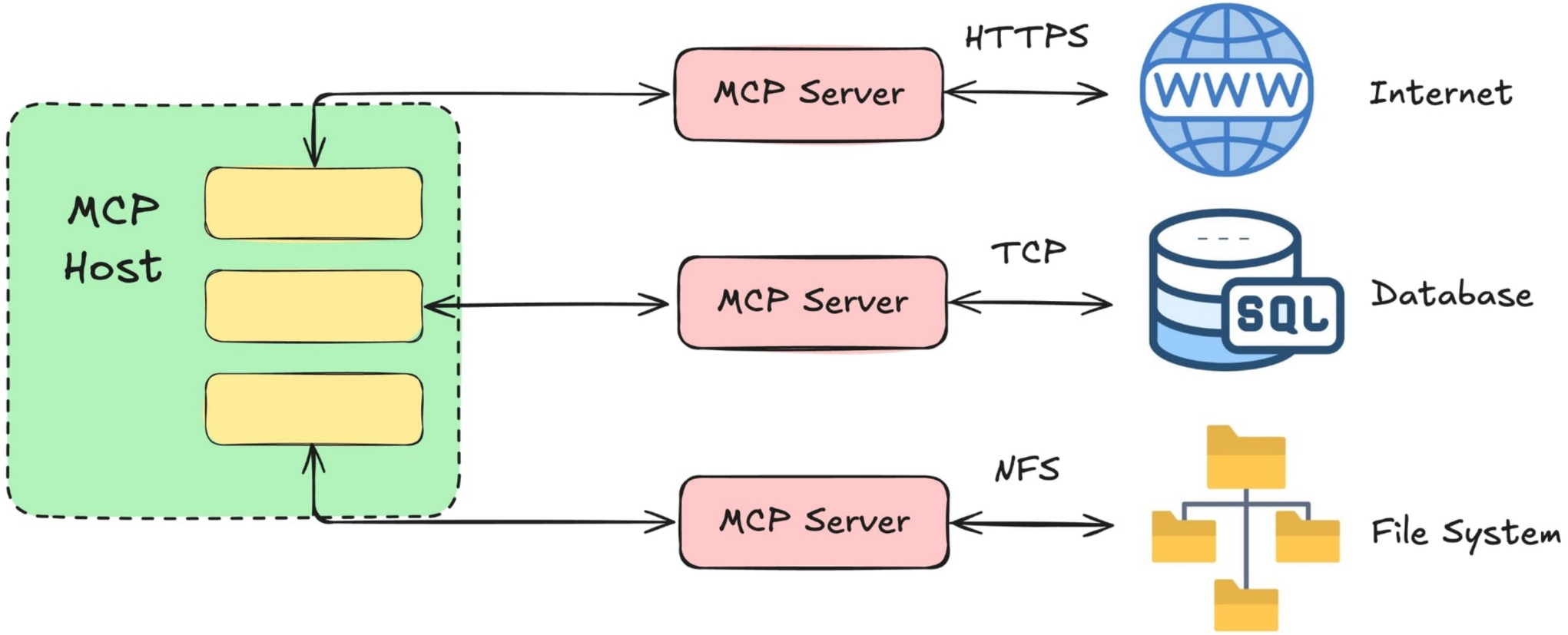

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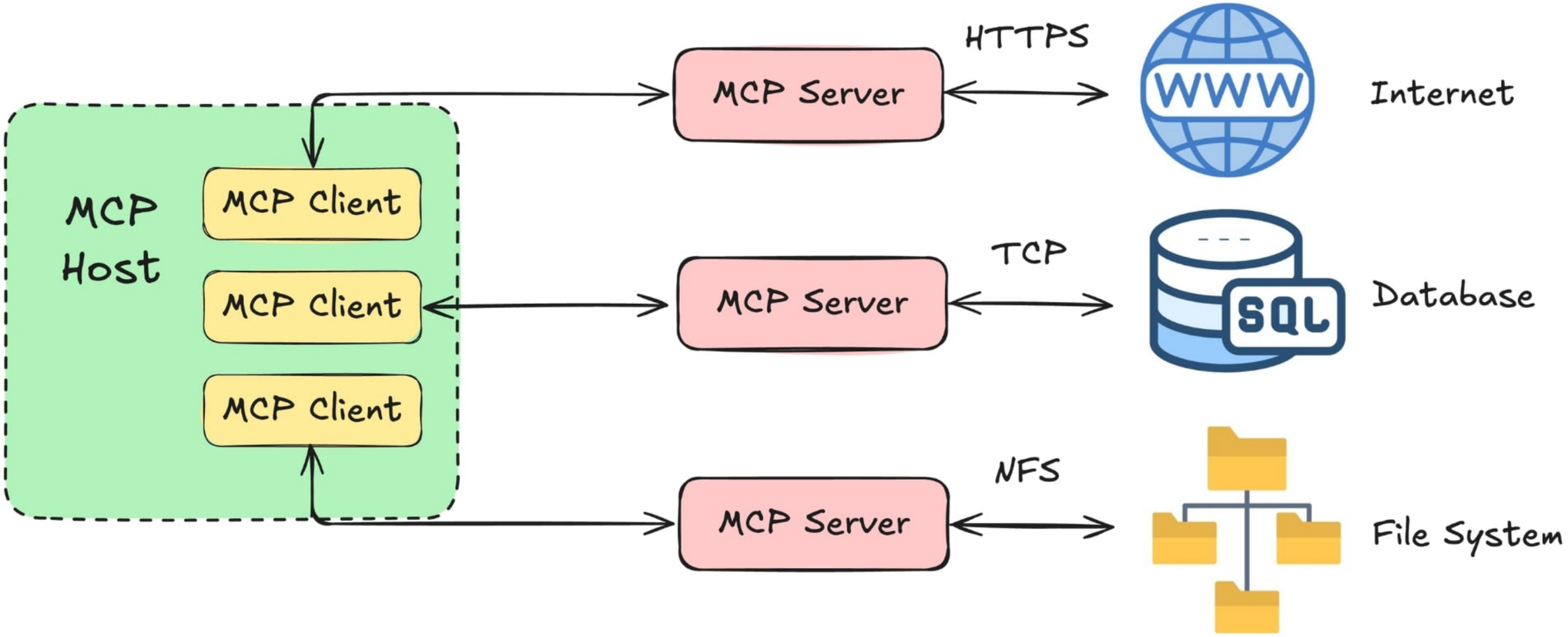

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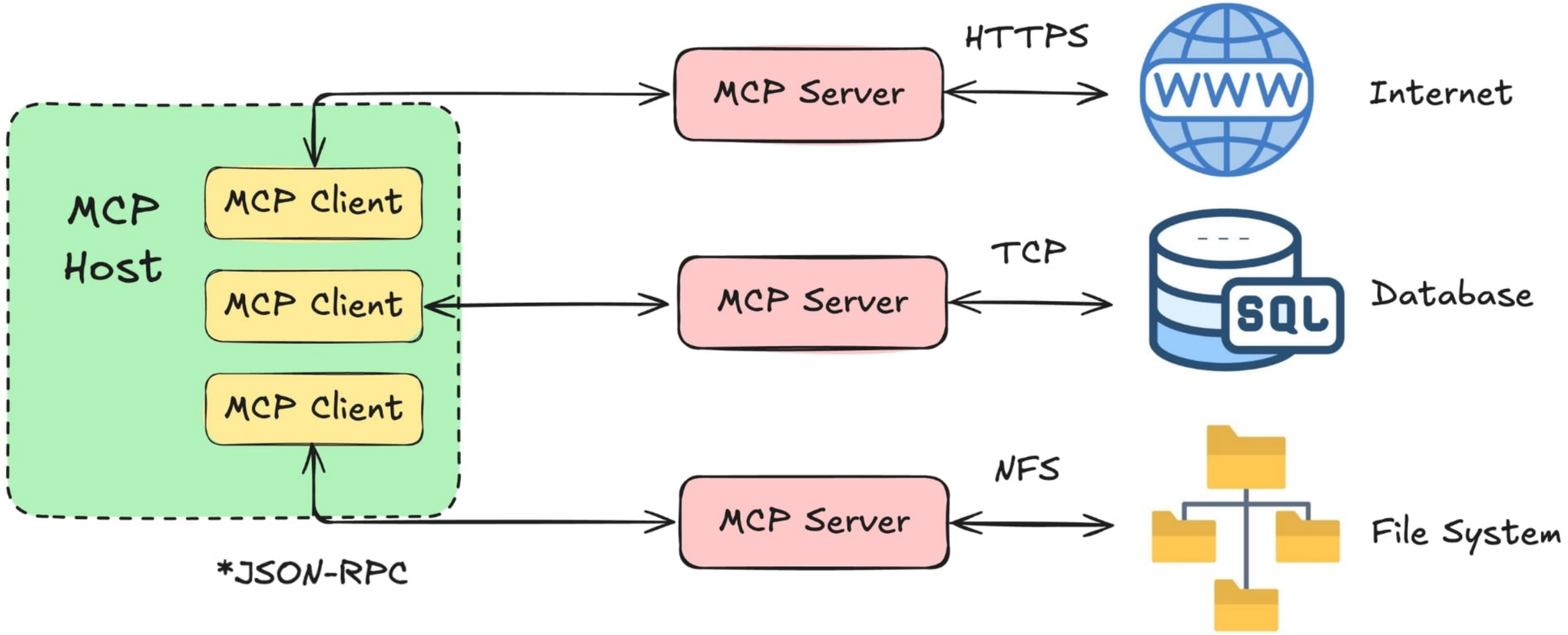

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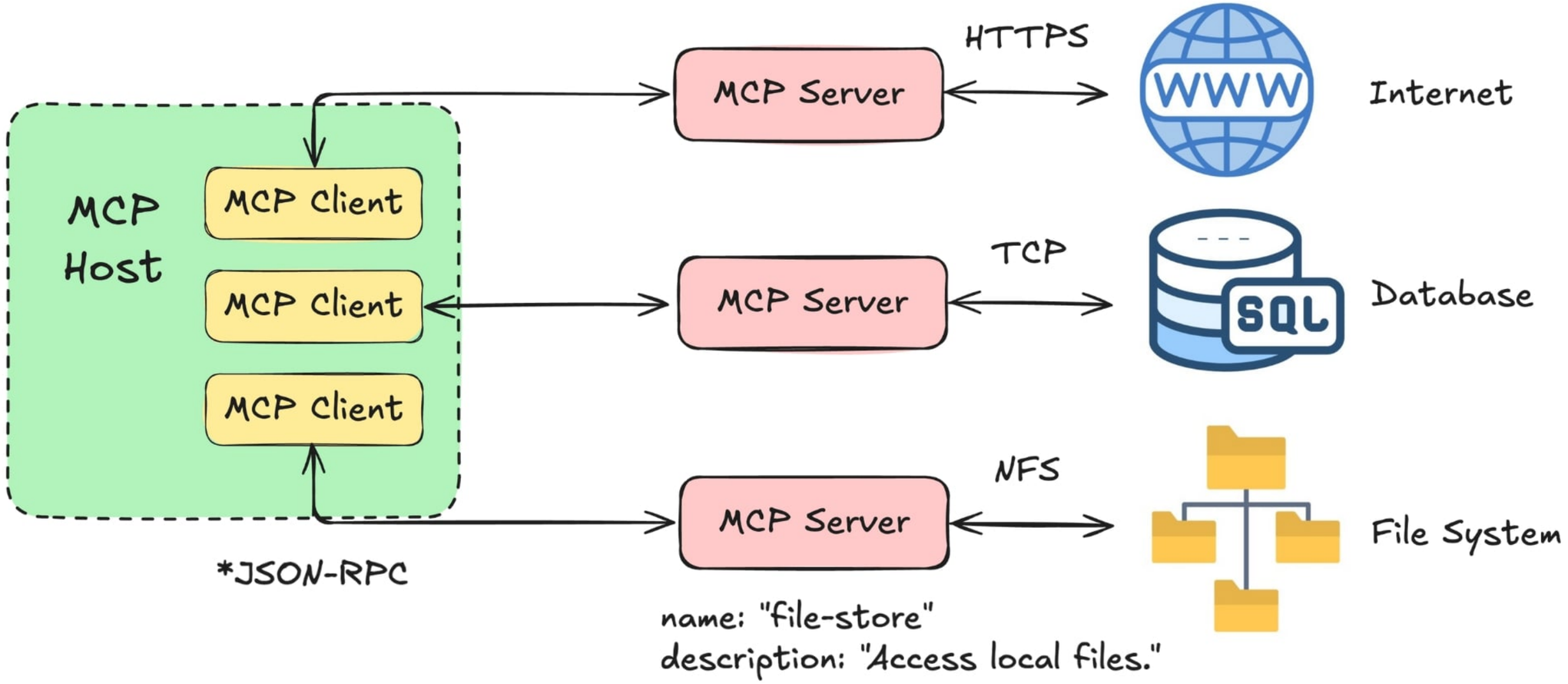
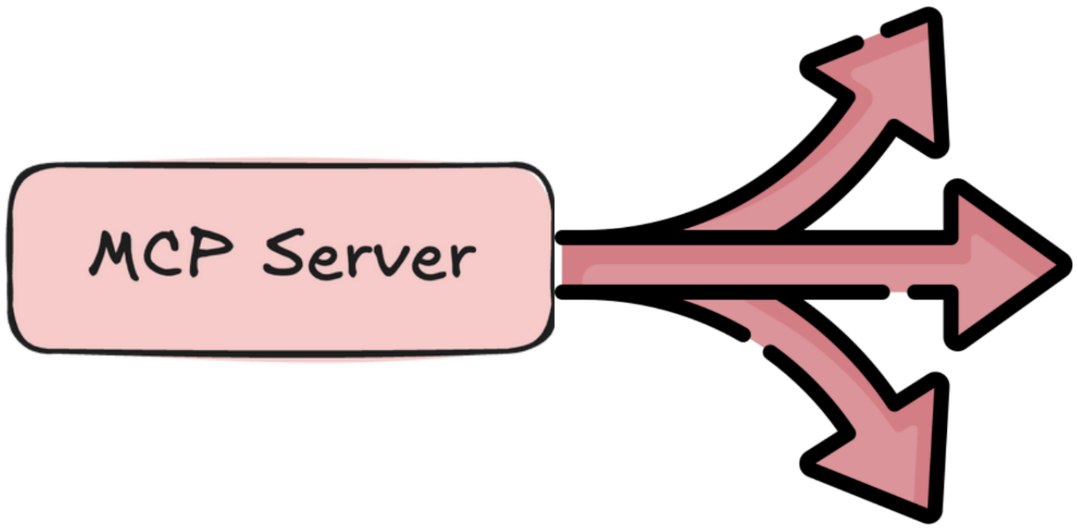


Primitives



Tools

-> Actions the LLM can perform
-> Querying a database, booking a flight, running code

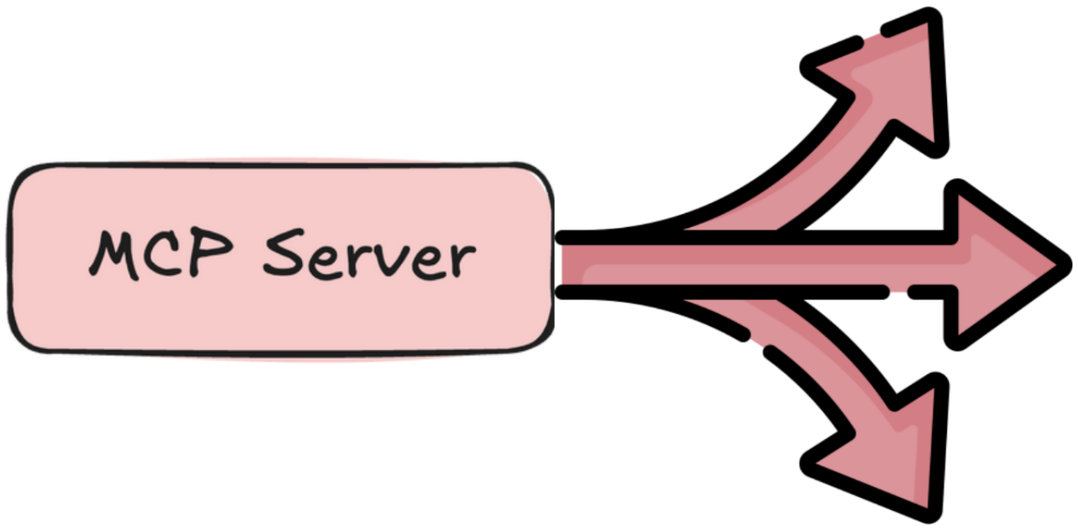
Resources

-> Read-only data or context
-> Documents, files, database records

Prompts

-> Pre-defined workflows and instructions
-> Saves users from needing to write long complex prompts

Primitives



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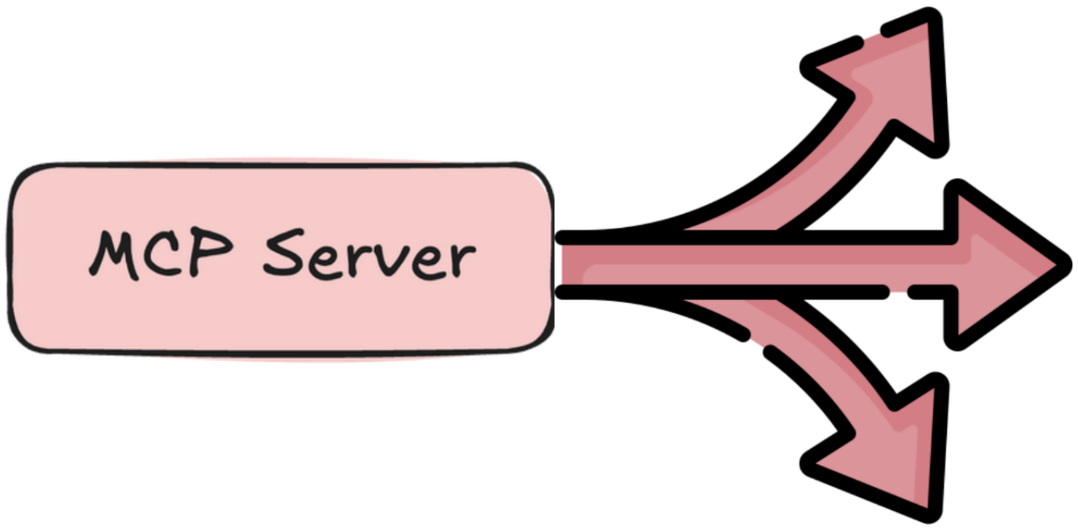

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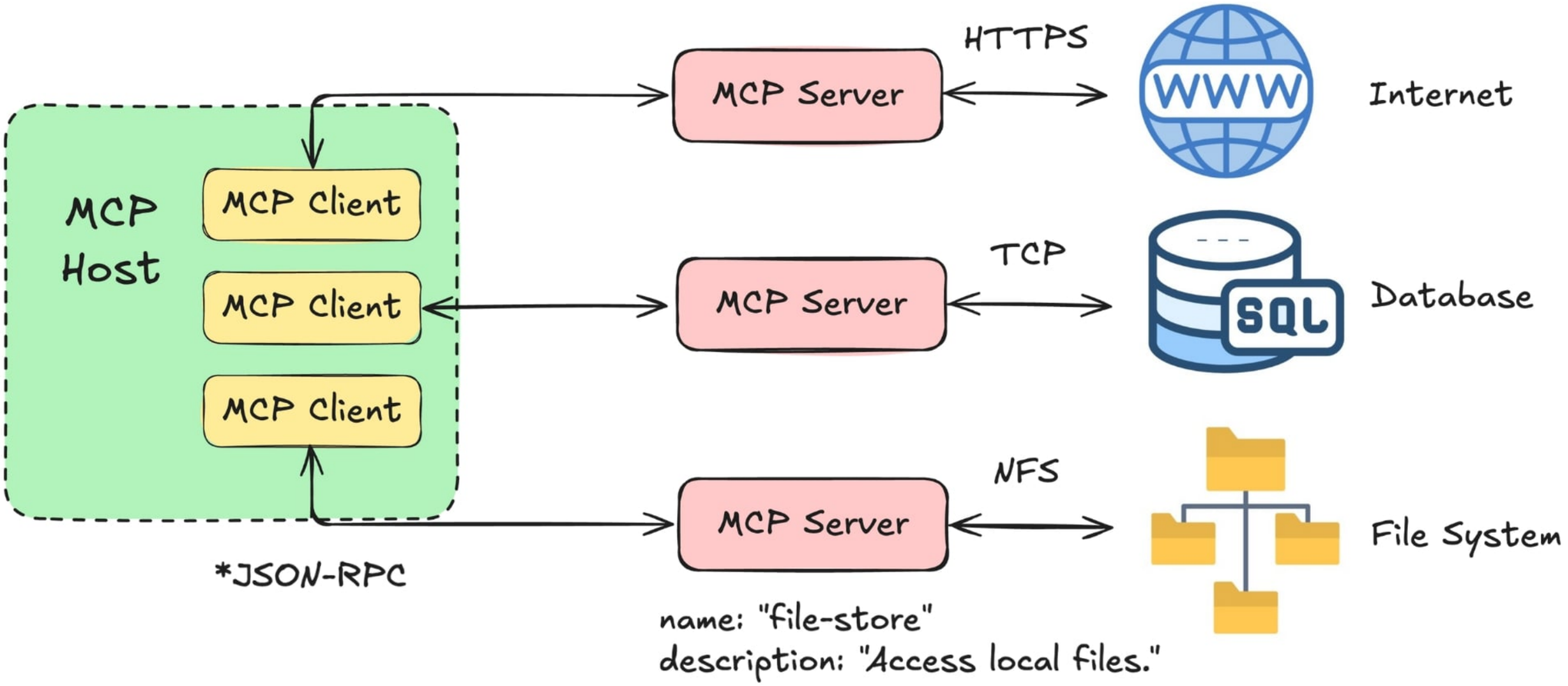

Let's practice!

INTRODUCTION TO MODEL CONTEXT PROTOCOL (MCP)

Building Your First MCP Server

INTRODUCTION TO MODEL CONTEXT PROTOCOL (MCP)

James Chapman
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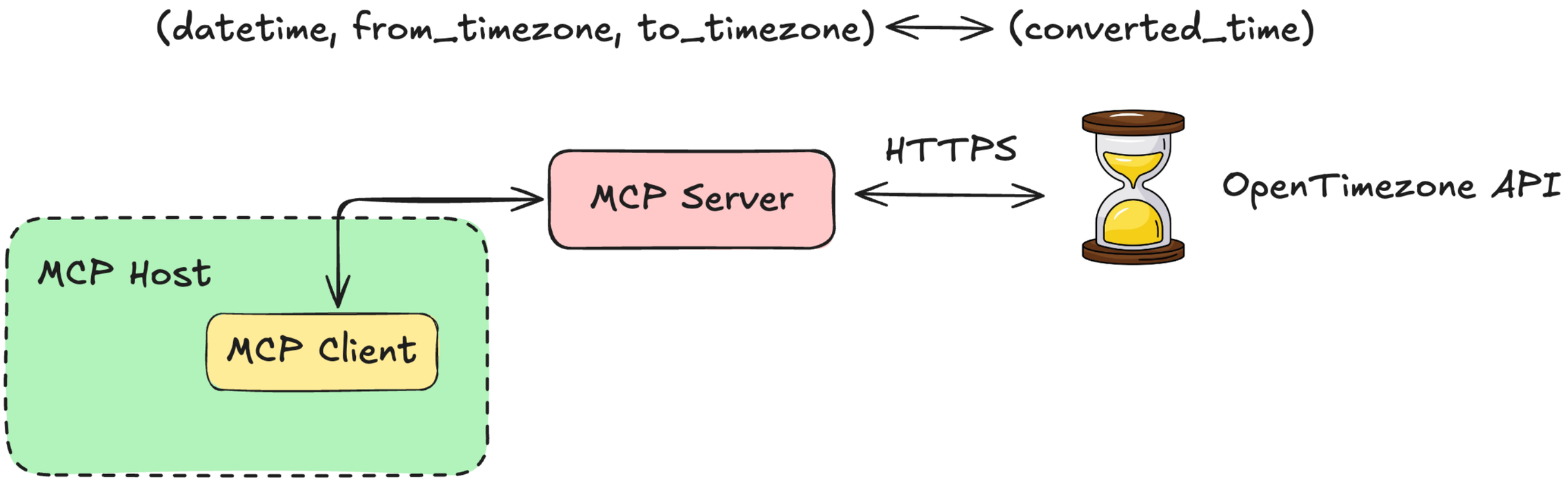
Converting Between Timezones



[1] https://opentimezone.com/

```python
# Create an MCP server instance
mcp = FastMCP("Timezone and Currency Converter")

@mcp.tool()
def convert_timezone(date_time: str, from_timezone: str, to_timezone: str) -> str:
    """
    Convert a datetime from one timezone to another.

    Args:
        date_time: The datetime string in ISO format (e.g., '2025-01-20T14:30:00')
        from_timezone: Source timezone (e.g., 'America/New_York')
        to_timezone: Target timezone (e.g., 'Europe/London')

    Returns:
        A string with the converted datetime and timezone information
    """
    # Define the API endpoint and format input data
    url = "https://api.opentimezone.com/convert"

    payload = {
        "dateTime": date_time,
        "fromTimezone": from_timezone,
        "toTimezone": to_timezone
    }

    # Make the API request
    response = requests.post(url, json=payload)

    data = response.json()
    converted_time = data.get('dateTime', 'N/A')

    return f"Time in {to_timezone}: {converted_time}"
```

Instantiate MCP Server

Define Tools

Doctrings and Typing

Step 1: Instantiating the Server

```python
from mcp.server.fastmcp import FastMCP
import requests

# Create an MCP server instance
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```

[1] https://github.com/modelcontextprotocol/python-sdk

Step 2: Adding the Tools

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    data = response.json()
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    return f"Time in {to_timezone}: {converted_time}"
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Step 3: Docstrings and Type Hints

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MCP Client-Server Communications

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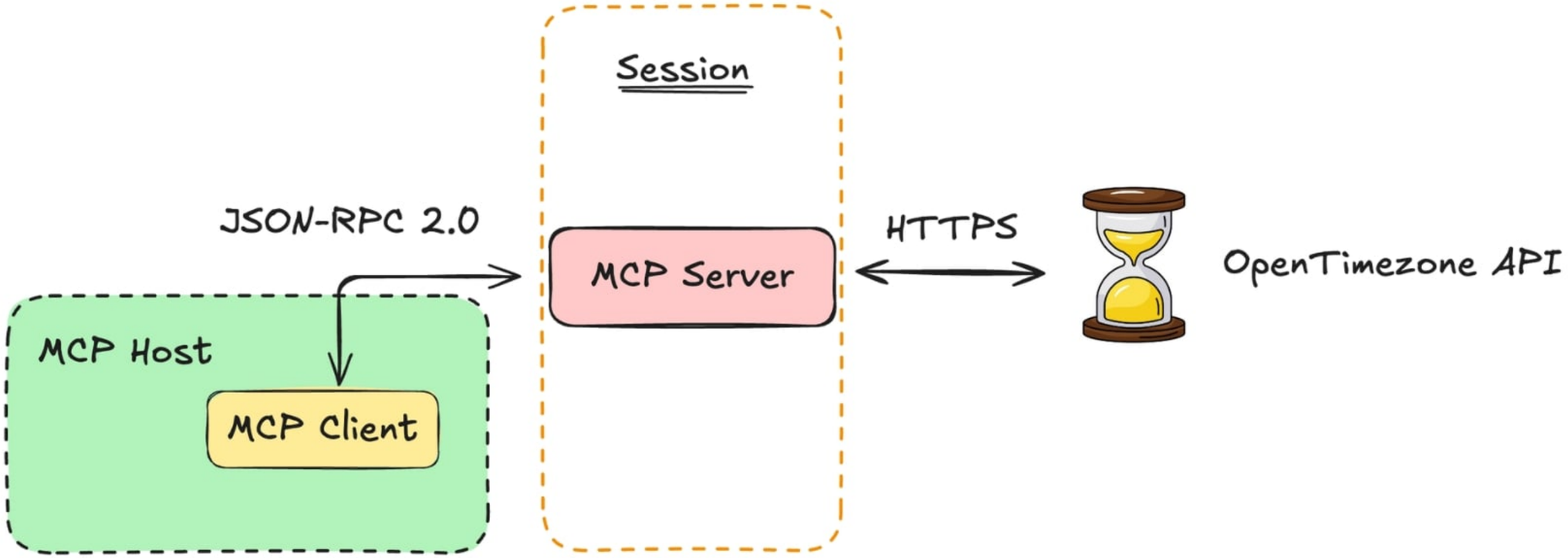
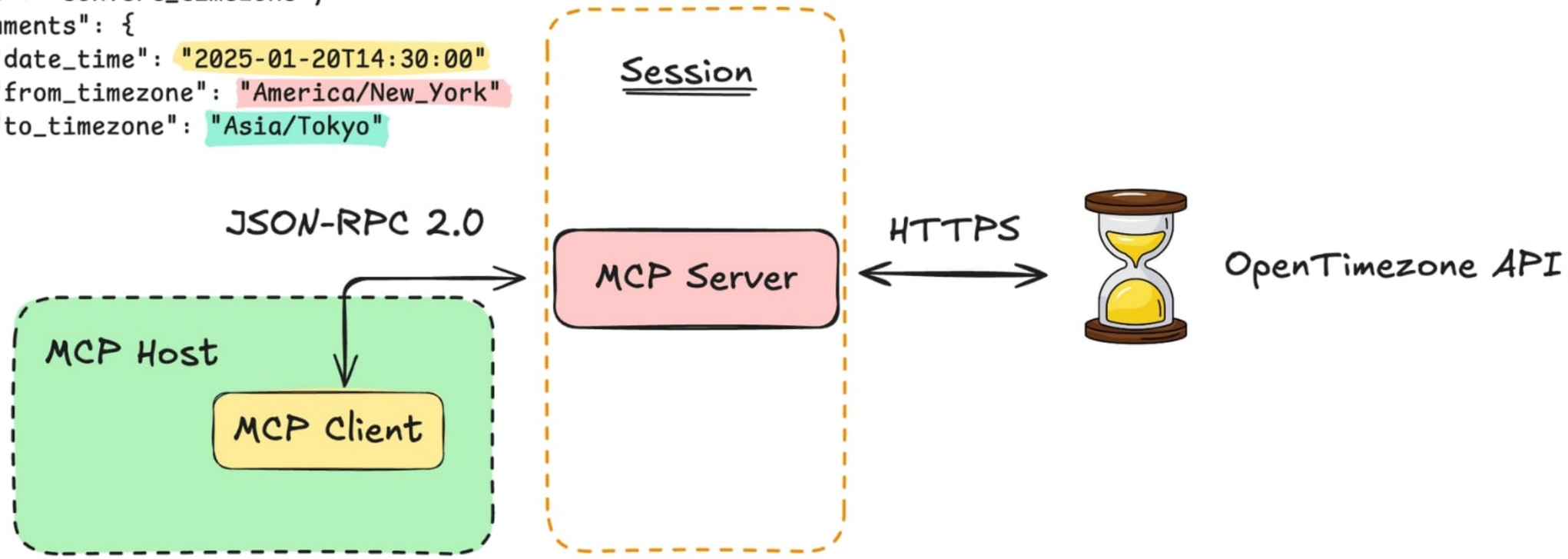
Client-Server Communications

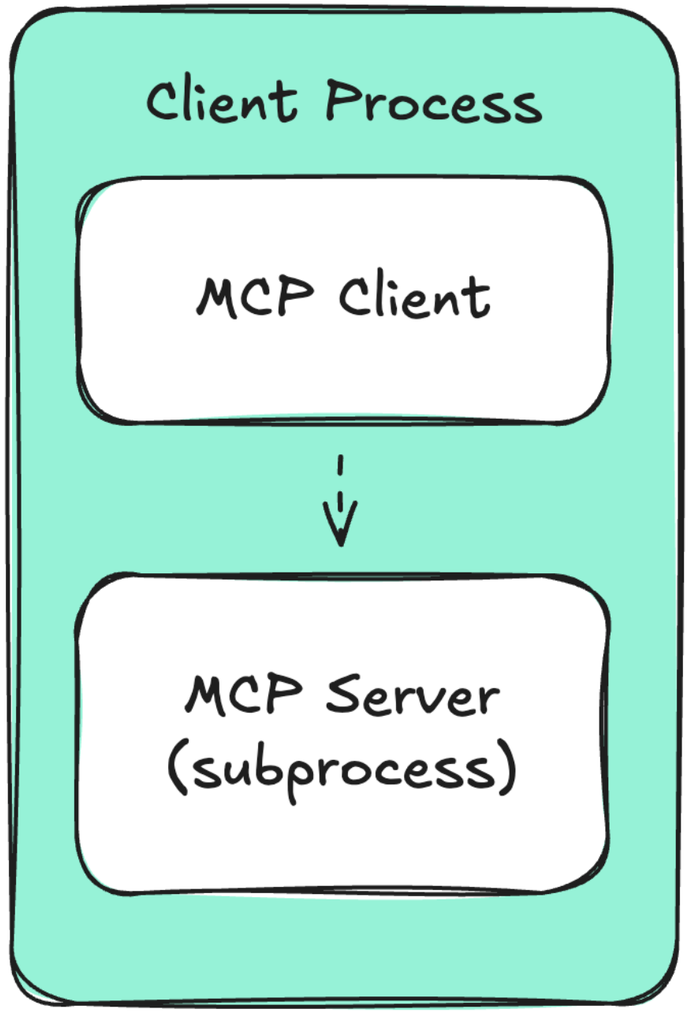


Client-Server Communications

```
{
    "jsonrpc": "2.0",
    "id": 1,
    "method": "tools/call",
    "params": {
        "name": "convert_timezone",
        "arguments": {
            "date_time": "2025-01-20T14:30:00"
            "from_timezone": "America/New_York"
            "to_timezone": "Asia/Tokyo"
        }
    }
}
```



Transports: Standard I/O (stdio)



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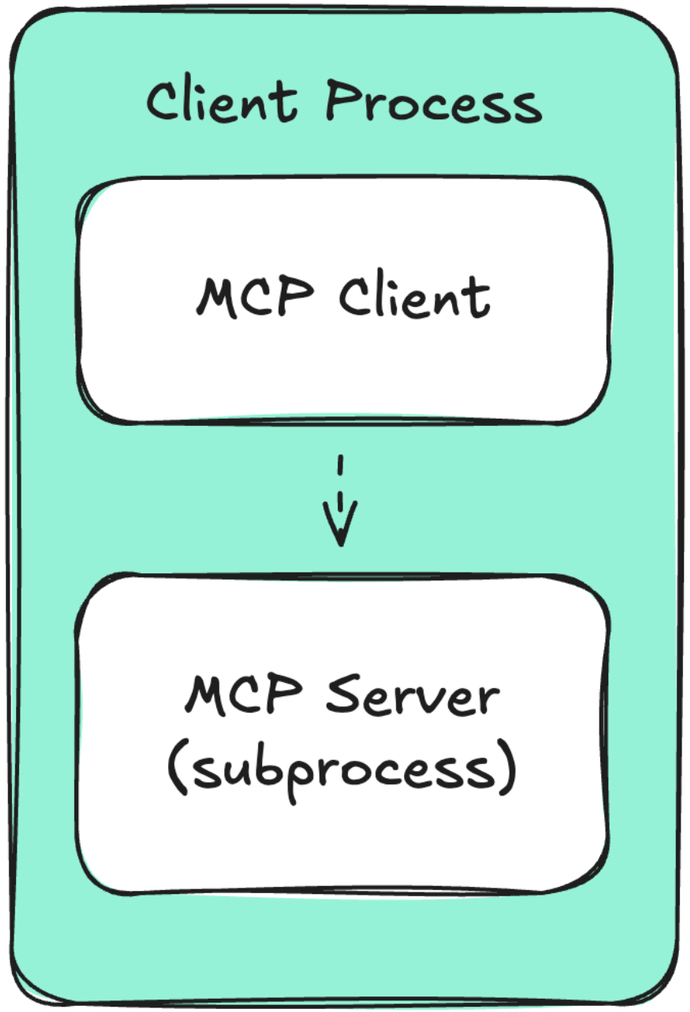
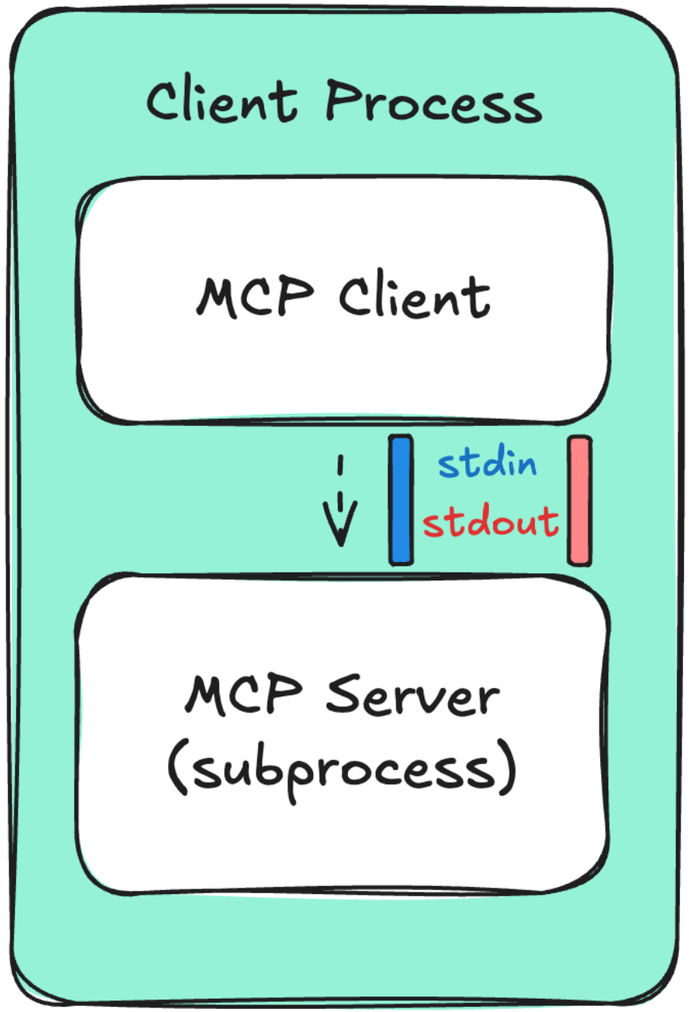


- Server runs as **child process**

Transports: Standard I/O (stdio)



- Server runs as **child process**
- Communication via pipes (**stdin/stdout**)

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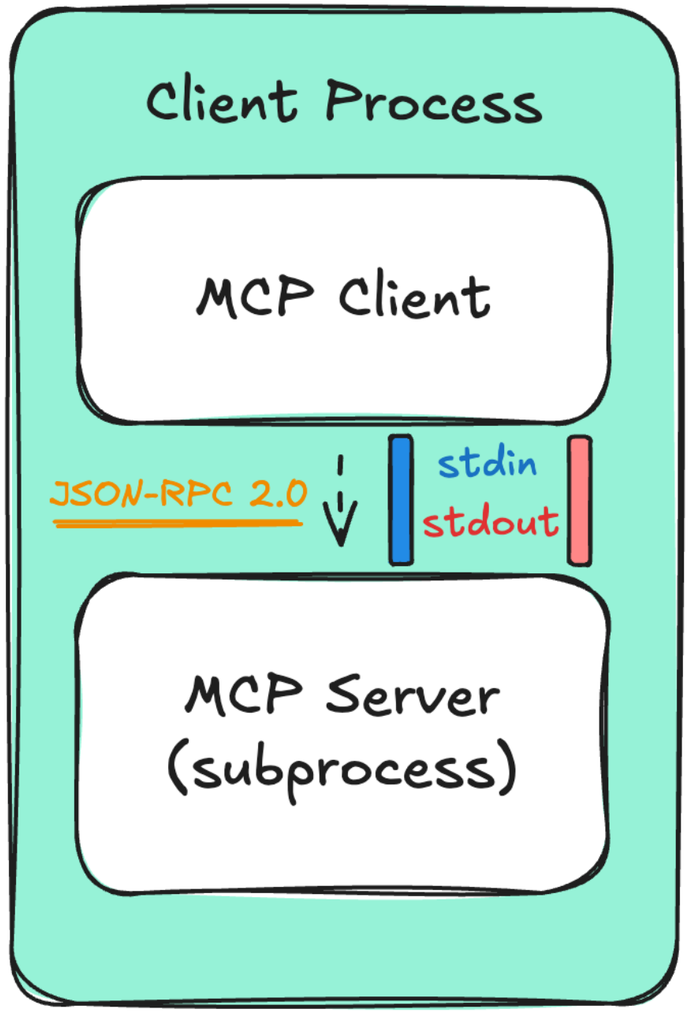

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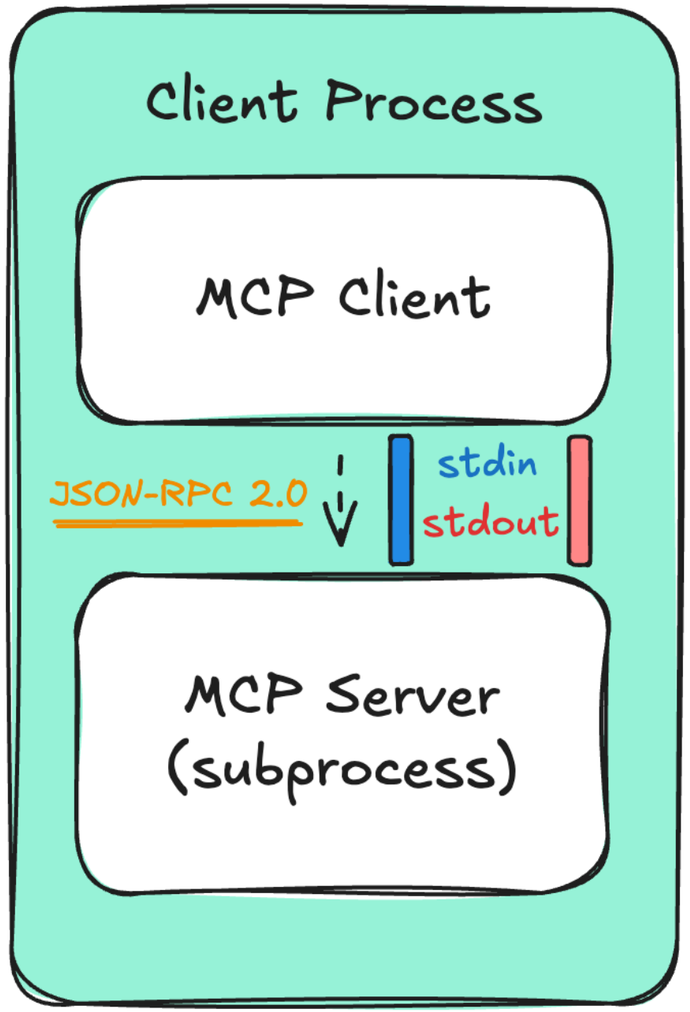
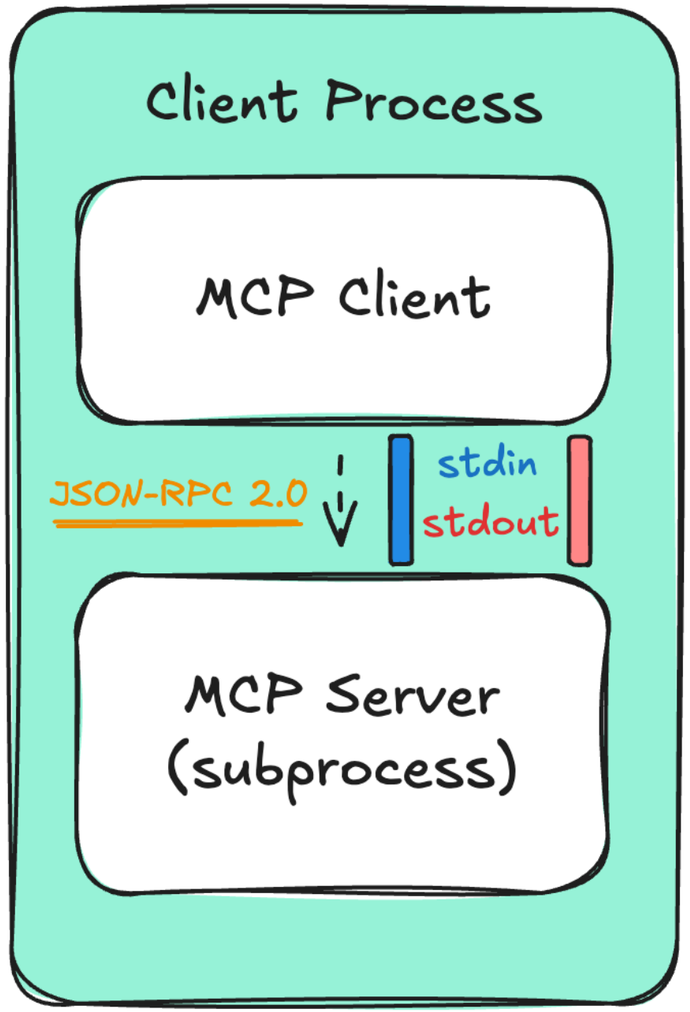
Transports: Standard I/O (stdio)



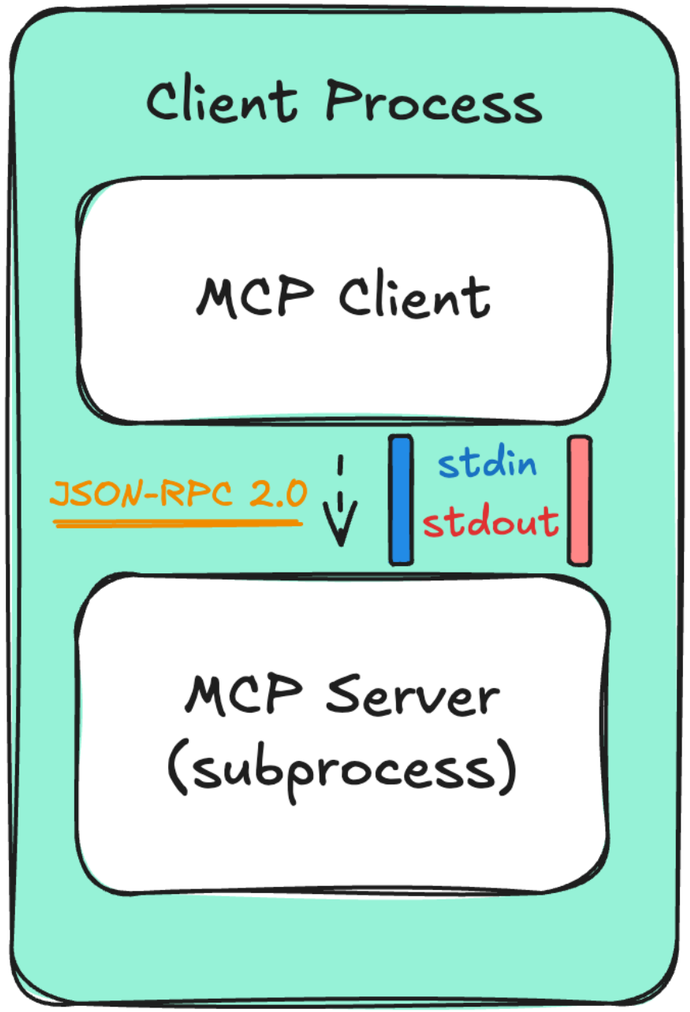
- Server runs as **child process**
- Communication via pipes (**stdin/stdout**)
- *Local only* - same machine

Transports: Standard I/O (stdio)



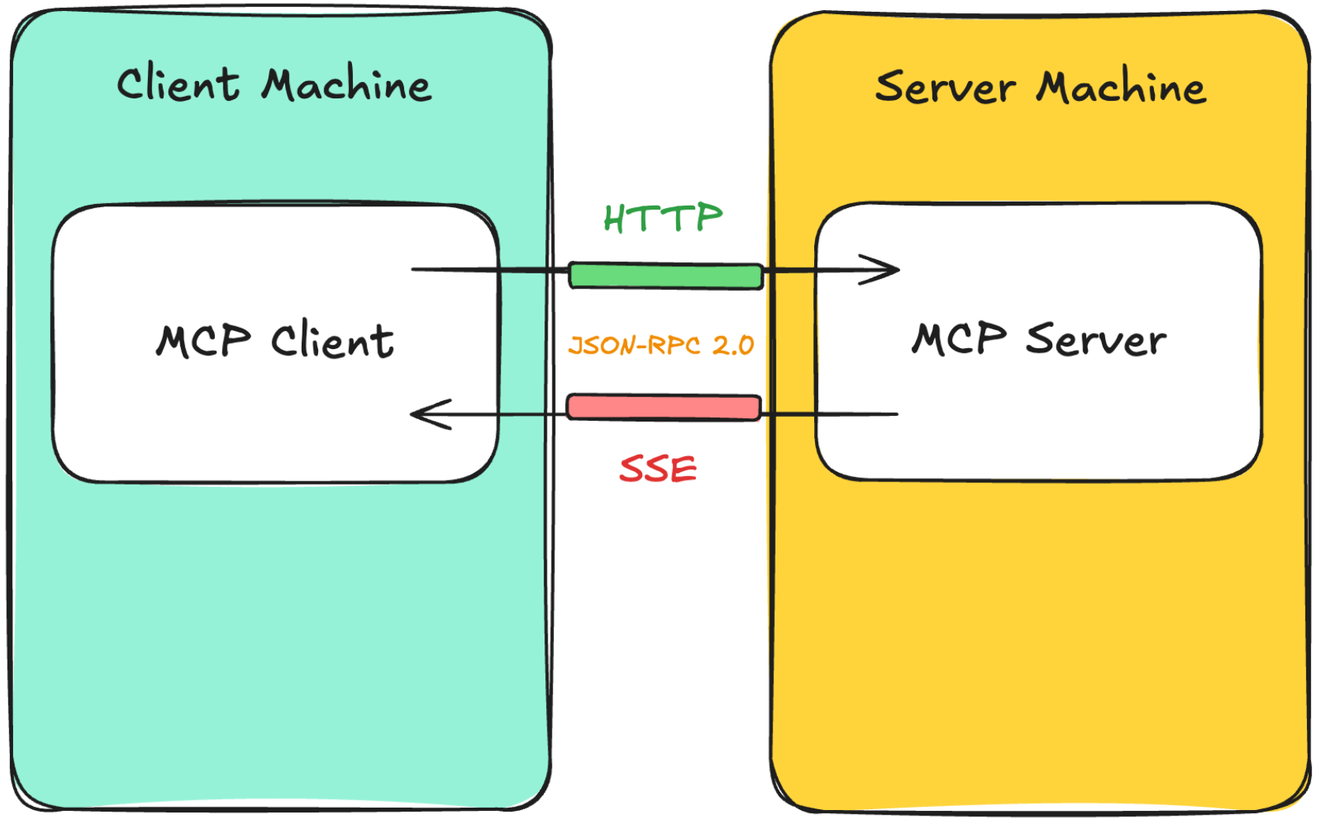
- Server runs as **child process**
- Communication via pipes (**stdin/stdout**)
- *Local only* - same machine
- Simple, no network configuration

Transports: Standard I/O (stdio)



- Server runs as **child process**
- Communication via pipes (**stdin/stdout**)
- *Local only* - same machine
- Simple, no network configuration
- **Server dies when client exits**

Transports: Streamable HTTP



MCP Server: timezone_server.py

```python
import mcp
from mcp.server.fastmcp import FastMCP
import requests

mcp = FastMCP("Timezone Converter")

@mcp.tool()
def convert_timezone(date_time: str, from_timezone: str, to_timezone: str) -> str:
    ...

if __name__ == "__main__":
    mcp.run(transport="stdio")
```

MCP Client: Listing Server Tools

```python
from mcp import ClientSession, StdioServerParameters
from mcp.client.stdio import stdio_client

async def get_tools_from_mcp():
    # Define the server parameters
    params = StdioServerParameters(command=sys.executable, args=["timezone_server.py"])

    # Connect to the MCP server and open a session
    async with stdio_client(params) as (reader, writer):
        async with ClientSession(reader, writer) as session:
            # Initialize the session
            await session.initialize()
            # Ask the server what tools it provides
            response = await session.list_tools()
```

MCP Client: Listing Server Tools

```
async def get_tools_from_mcp():
            # ...

            # Ask the server what tools it provides
            response = await session.list_tools()
            print("Connected to MCP server!")
            print("Available tools:")
            for tool in response.tools:
                print(f" - {tool.name}: {tool.description}")

            return response.tools

asyncio.run(get_tools_from_mcp())
```

MCP Client: Listing Server Tools

```
Connected to MCP server!
Available tools:
 - convert_timezone:
    Convert a datetime from one timezone to another.

    Args:
        date_time: The datetime string in ISO format (e.g., '2025-01-20T14:30:00')
        from_timezone: Source timezone (e.g., 'America/New_York')
        to_timezone: Target timezone (e.g., 'Europe/London')

    Returns:
        A string with the converted datetime and timezone information
```

MCP Client: Calling Server Tools

```python
async def call_mcp_tool(tool_name: str, arguments: dict) -> str:
    params = StdioServerParameters(command=sys.executable, args=["timezone_server.py"])

    async with stdio_client(params) as (reader, writer):
        async with ClientSession(reader, writer) as session:

            await session.initialize()

            result = await session.call_tool(tool_name, arguments)

            text_content = result.content[0].text

            print(f"Conversion Result: {text_content}")
            return str(text_content)
```

MCP Client: Calling Server Tools

```
asyncio.run(
    call_mcp_tool(
        "convert_timezone",
        {"date_time": "2025-01-20T14:30:00",
         "from_timezone": "America/New_York",
         "to_timezone": "Asia/Tokyo"}
    )
)
```

```
'2025-01-21T04:30:00+09:00'
```

Let's practice!

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